

Osteomyelitis & Skin Infections – Summary

Key Symptoms:

- **Osteomyelitis:** Pain, fever, tender, swelling, ulcer formation (vascular insufficiency), limited motion
- **Skin:** Warm/hot, erythema, tender, weeping/crusting, itchy

Most common causing-pathogens:

- **Hematogenous:** *S. aureus*, streptococci, pseudomonas (people who inject drugs), salmonella (sickle cell anaemia) or tuberculosis
- **Contiguous:** *S. aureus*, pseudomonas, proteus, klebsiella, *E. coli* or *B. fragilis* (anaerobic)
- **Vascular insufficiency:** *S. aureus*, β -hemolytic strep, coagulase-negative staph, gram negative and anaerobes. Usually polymicrobial.
- **Uncomplicated SSTIs:** e.g., cellulitis/erysipelas, impetigo, ecthyma, folliculitis, etc.
 - *S. aureus*, β -hemolytic strep (groups A, B, C, G) and pyogenes
 - **Fresh water:** *Aeromonas hydrophila*
 - **Salt water:** *Vibrio vulnificus*
 - **Soil:** *Pseudomonas* spp., *Clostridium* spp., *Bacillus* spp.
 - **Facial cellulitis:** *Haemophilus influenzae*.
- **Complicated SSTIs:** e.g., necrotizing fasciitis/cellulitis, animal bite, diabetic ulcer, etc.
 - *S. aureus*, enterococcus spp., *P. aeruginosa*, *E. coli* and other Enterobacteriaceae, environmental and zoonotic pathogens
 - **Animal bites:** *S. aureus*, strep and **mouth anaerobes** (*Pasteurella* spp., *Elkenella* spp.)

Non-Pharmacological Treatment: Surgical drainage/debridement and wound care

Pharmacological Treatment: Typical 4-8 weeks therapy for osteomyelitis, typical 5-7 days for SSTI

- **Hematogenous:** Cefazolin, 3rd/4th gen cephalosporins, pip-tazo, carbapenems, MRSA coverage prn
- **Contiguous:** Cefazolin, 3rd/4th gen cephalosporins, pip-tazo, carbapenems, MRSA coverage prn
- **Vascular insufficiency:** 3rd/4th gen cephalosporins, pip-tazo, carbapenems, metronidazole, clindamycin, MRSA coverage prn
 - MRSA coverage: vancomycin, linezolid or daptomycin
- **Uncomplicated SSTIs:**
 - **Topical Abx:** Mupirocin, fusidic acid
 - **Oral Abx:** Amoxi-clav, cephalexin, clindamycin, cloxacillin, doxycycline, TMP/SMX
- **Complicated SSTIs:**
 - **Oral:** Amoxi-clav, cephalexin, clindamycin, cloxacillin, doxycycline, moxifloxacin
 - **IV:** Amoxi-clav, pip-tazo, carbapenems, vancomycin, 3rd gen cephalosporins

Monitoring Parameters: Check if pt is responding to IV therapy -> blood panel, TDM if required (vancomycin or aminoglycosides)

Additional Key Points:

- **Anti-pseudomonal antibiotics:** ceftazidime, pip-tazo, cefepime, carbapenems (except ertapenem)
- For vascular insufficiency in diabetics, consider anaerobic coverage: e.g., metronidazole, clindamycin, pip-tazo, etc.
- Empiric therapy is not needed unless there is sepsis or an inability to get cultures
- Antibiotics do not penetrate well into areas of high amounts of pus or into bone that has a compromised blood supply, therefore, a combination of surgical and an antimicrobial approach should be considered.
- If purulent (pus) = Staph, if non-purulent = Strep